

REFERME INVESTMENTS

DATA QUALITY RULES

Investment Analytics & Decision-Support Framework

February - June 2026

Document Control	Details
Document ID	RMI-DQ-001
Version	1.0
Purpose	Define the source-supported data-quality validation rules used to protect KPI calculation, scoring confidence and investment decision support.
Requirement Link	BR-10: Data quality shall be measured as part of the risk framework; FR-12: expose data-quality status.
Primary UAT Link	UT-09: Remove/invalid required inputs -> data-quality status declines and limitation is visible.
Source Discipline	Rules DQ-01 to DQ-08 are taken from the existing project artifacts. Final completeness thresholds and detailed source-specific validation logic remain subject to approval.

1. Purpose

Data quality is part of the ReferMe Investments risk framework because incomplete, invalid, duplicate or stale evidence can reduce confidence in an investment evaluation. These rules define the validation controls that should be applied before KPI results, scores and recommendations are treated as decision-ready.

2. Data Quality Principles

Missing or low-quality data should reduce confidence in the recommendation and remain visible to decision makers.

A high composite score should not be treated as conclusive when mandatory evidence is missing or data quality is weak.

Data-quality validation occurs before the 28-KPI calculation and scoring stages in the proposed TO-BE process.

Data-quality results contribute to the Risk dimension through the Data Quality Score / status.

The exact approved completeness threshold is not established in the source and must not be invented.

3. Data Quality Rule Catalogue

Rule ID	Rule	Domain	Failure Treatment	Primary Data Element(s)	Quality Dimension
DQ-01	Business ID must be present and valid.	Business Master	Reject record / return for correction.	Business ID	Identity / mandatory field
DQ-02	Evaluation period must be present and valid.	Time	Reject period-based KPI calculation.	Evaluation period / time fields	Period validity
DQ-03	Revenue denominators used in margin calculations must be non-zero.	Financial	Return KPI as not calculable; flag data issue.	Revenue	Calculation validity
DQ-04	Percentage KPIs must fall within logically valid bounds	Derived KPI / multiple domains	Flag exception for analyst review.	Percentage KPI outputs	Range / validity

	unless business definition permits otherwise.				
DQ-05	Duplicate BusinessID + evaluation-period records must be resolved before scoring.	Business Master + Time	Block duplicate contribution to score.	Business ID + evaluation period	Uniqueness / duplicate control
DQ-06	Mandatory financial, operational and customer inputs must meet approved completeness threshold.	Financial / Operations / Customers	Lower confidence / Data Quality Score.	Mandatory evaluation inputs	Completeness
DQ-07	KPI source period must align with the evaluation period.	All period-based source data	Flag stale or mismatched evidence.	Source period vs evaluation period	Timeliness / consistency
DQ-08	Scoring weights must total 100% before Investment Score calculation.	Scoring configuration	Block scoring configuration.	Dimension scoring weights	Configuration integrity

4. Rule Application in the TO-BE Process

Process Stage	Data Quality Control	Outcome
Standardized data intake	Check Business ID, evaluation period and required inputs.	Valid records proceed; invalid/insufficient records are returned, blocked or flagged according to the applicable rule.
Data-quality validation	Apply DQ-01 to DQ-08 as applicable.	Data-quality status/confidence is established.
KPI calculation	Protect denominators, ranges and period alignment.	KPIs calculate only when source evidence is valid enough for that calculation.
Scoring	Prevent duplicate contribution and invalid scoring configuration.	Dimension/Investment Score is calculated only under valid scoring setup.
Decision support	Expose Data Quality Score/status beside the evaluation.	Decision maker can distinguish weak performance from weak evidence.
Monitoring / alerts	Use current, period-aligned evidence for trend and risk views.	Stale or mismatched evidence is visible rather than silently interpreted.

5. Data Element Validation Mapping

Source Domain / Logical Field	Transformation / Business Logic	Target KPI / Model Field	Validation	DQ Rule Link
Business Master.BusinessID	Trim; mandatory; unique business identifier	DimBusiness[BusinessID]	Not null; unique in dimension	DQ-01, DQ-05
Business Master.BusinessType	Standardize business-type labels	DimBusiness[BusinessType]	Allowed standardized values	Source validation requirement; detailed allowed list TBD
Financial.Revenue	Aggregate by BusinessID + period	[Revenue]	Numeric; >=0; reconcile to source	DQ-03, DQ-07
Financial.COGS	Revenue - COGS	[Gross Profit] -> [Gross Margin %]	Numeric; period aligned	DQ-03, DQ-07
Financial.EBITDA	Aggregate by period; EBITDA / Revenue	[EBITDA], [EBITDA Margin %]	Numeric; denominator check	DQ-03, DQ-07
Financial.OperatingCashFlow	Aggregate; OCF / Revenue	[Operating Cash Flow], [Cash Flow Margin %]	Numeric; period aligned	DQ-03, DQ-07
Operations.EmployeeCount	Revenue / average employee count	[Revenue per Employee]	Positive headcount	DQ-04 / source validation
Operations.CapacityUsed + CapacityAvailable	Used / available capacity	[Capacity Utilization %]	Available > 0	DQ-04
Operations.OnTimeServices + TotalServices	On-time / total services	[On-Time Service %]	Total > 0	DQ-04
Customers.ActiveCustomers	Compare current vs prior approved period	[Customer Growth %]	Period alignment	DQ-07
Customers.Retained / Eligible	Retained / eligible prior customers	[Customer Retention %]	0-100%	DQ-04

Customers.Lost / Eligible	Lost / eligible prior customers	[Customer Churn %]	0-100%	DQ-04
Customers.RepeatCustomers	Repeat / total customers	[Repeat Customer Rate]	0-100%	DQ-04
Customers.TopCustomerRevenue	Top customer(s) revenue / Revenue	[Customer Concentration %]	0-100%; definition approved	DQ-03, DQ-04
Financial.Revenue + Date	Approved 12-month growth methodology	[12-Month Revenue CAGR]	Requires sufficient history	DQ-02, DQ-07
Financial.Revenue + Date	Calculate variability over approved lookback	[Revenue Volatility]	Requires history	DQ-02, DQ-07
Data validation results	Composite completeness/validity/consistency result	[Data Quality Score]	Rules approved	DQ-01 to DQ-07
Five dimension scores	Weighted sum using approved dimension weights	[Investment Score]	Weights total 100%	DQ-08

6. Data Quality Score / Confidence

The project establishes Data Quality Score as one of the six Risk KPIs and defines it as a composite completeness / validity / consistency result. The exact production formula, weighting of quality dimensions, and numeric cut-offs are not established in the source. Accordingly, this document does not invent a formula or score band.

Aspect	Source-Supported Position
Purpose	Make confidence in the source evidence visible to decision makers.
Inputs	Data validation results, including missing, invalid, duplicate or stale/mismatched indicators.
Risk treatment	Data Quality Score/status is included in the Risk dimension and evaluation confidence.
Low-quality evidence	Should lower confidence and remain visible; mandatory completeness must meet an approved threshold.
Formula / weights	TBD / requires approval.
Thresholds	TBD / requires approval.

7. Exception Handling

Exception	Required Treatment
Missing / invalid Business ID	Reject record or return for correction.
Missing / invalid evaluation period	Reject period-based KPI calculation.
Zero revenue denominator for margin calculation	Return affected KPI as not calculable and flag the data issue.
Percentage outside logically valid bounds	Flag exception for analyst review unless the approved business definition permits it.
Duplicate BusinessID + period	Resolve before scoring; duplicate must not contribute to score.
Insufficient mandatory inputs	Lower confidence / Data Quality Score according to approved completeness threshold.
Period mismatch / stale evidence	Flag stale or mismatched evidence.

Weights not totaling 100%	Block Investment Score calculation/configuration.
----------------------------------	---

8. Requirements and UAT Traceability

Trace ID	Traceability
BR-10	Data quality shall be measured as part of the risk framework.
FR-12	Expose data-quality status.
US-05	Decision maker wants data-quality visibility to distinguish weak performance from weak evidence.
UT-09	Remove/invalid required inputs -> data-quality status declines and limitation is visible.
Risk KPI	Data Quality Score - completeness/validity/consistency of required data.

9. Validation Ownership

The source-supported RACI assigns Data / BI as A/R for validating data-quality rules, with Investment / Underwriting, Finance, Operations and the Investment Target consulted as applicable.

Activity	Investment / Underwriting	Finance Analyst	Operations / Portfolio	Data / BI	Investment Target
Validate data quality rules	C	C	C	A/R	C

10. Open Items Requiring Approval

Approved completeness threshold for mandatory financial, operational and customer inputs.

Final allowed-value lists and source-specific validation constraints.

Definition of stale data / freshness threshold.

Final formula and weighting for Data Quality Score.

Business-approved exceptions where a percentage KPI may validly fall outside a default logical range.

Physical source-system controls, owners and operational remediation workflow.

